IEOR 221 - HW 2 - Jie Wen

1)

(a) Forward price and initial value

The forward price K for a non-dividend-paying stock is given by $K = S_0 e^{rT}$, where $S_0 = 40$ is the initial stock price, $r = 0.05$ is the risk-free rate, and $T = 1$ is the time to maturity.

Substitute the values:
$K = 40 \times e^{0.05 \times 1} \approx 42.05$.

The initial value of the forward contract is **0**, as the forward price is chosen such that the contract has zero value at inception.

(b) Forward price and value six months later

Six months later ($t = 0.5$), the remaining time to maturity is $T - t = 0.5$, the stock price is $S_t = 45$, and the risk-free rate is still $r = 0.05$.

The forward price (for a new contract initiated at $t = 0.5$) is $F = S_t e^{r(T-t)}$.
Substitute the values:
$F = 45 \times e^{0.05 \times 0.5} \approx 46.14$.

The value of the original forward contract at $t = 0.5$ is $f_t = S_t - Ke^{-r(T-t)}$, where $K \approx 42.05$ is the original forward price.
First, compute $Ke^{-r(T-t)} = 42.05 \times e^{-0.025} \approx 41.01$.
Then, $f_t = 45 - 41.01 \approx 3.99$.

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2)

(a) Forward price and initial value

The forward price K is given by $K = (S_0 - PV(D))e^{rT}$, where $S_0 = 40$, $r = 0.05$, $T = 1$, and $PV(D)$ is the present value of the dividends.

The dividends are $3at\$t = 0.25$ and $3at\$t = 0.75$.

$PV(D) = 3e^{-0.05 \times 0.25} + 3e^{-0.05 \times 0.75} \approx 5.85$.

Thus, $K \approx (40 - 5.85) \times e^{0.05} \approx 35.90$.

The initial value of the forward contract is **0**.

(b) Forward price and value six months later

Six months later ($t = 0.5$), the stock price is $S_t = 45$, and the remaining time to maturity is 0.5 years.

The remaining dividend is $3at\$t = 0.75$ (0.25 years from now).

$PV_t(D) = 3e^{-0.05 \times 0.25} \approx 2.96$.

The new forward price (for a contract initiated at $t = 0.5$) is $F = (S_t - PV_t(D))e^{r \times 0.5} \approx (45 - 2.96) \times e^{0.025} \approx 43.10$.

The value of the original forward contract is $f_t = S_t - PV_t(D) - Ke^{-r \times 0.5} \approx 45 - 2.96 - 35.90 \times e^{-0.025} \approx 7.03$.

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3)

The time to maturity is $T = 5/12$ years, assuming each month is $1/12$ of a year.

The integral of the dividend yield over the period is $\int_0^T q(t)\,dt = (0.05 + 0.02 + 0.02 + 0.05 + 0.02)/12 = 0.16/12$.

The exponent in the futures price formula is $rT - \int_0^T q(t)\,dt = 0.04 \times (5/12) - 0.16/12 = 1/300$.

The futures price is $F = 1300 \times e^{1/300}$.

To compute this numerically, $e^{1/300} \approx 1.003338$, so $F \approx 1300 \times 1.003338 = 1304.34$.

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4)

The futures contract is overpriced relative to the theoretical fair price of approximately 402.68.

To exploit this arbitrage opportunity:

- Borrow 394.70 at the risk-free rate of 6% per annum (continuous compounding) for four months.
- Use the borrowed funds to purchase $e^{-qT} \approx 0.9868$ units of the stock index at the current price of 400 (total cost: 394.70).
- Simultaneously, sell (short) the futures contract at the quoted price of 405.

At maturity in four months:

- With continuous dividend reinvestment at 4% per annum, the index position will grow to exactly 1 unit.
- Settle the futures contract by effectively delivering 1 unit of the index (cash settlement equivalent: receive 405 and pay the then-prevailing index value, but since you hold the equivalent position, it nets out).
- Repay the loan, which has grown to approximately 402.68.
- The risk-free profit is 405 - 402.68 = **2.32** (per unit of the index underlying the contract).

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5)

The theoretical forward exchange rate, based on covered interest parity, is $F = S \times e^{(r_d - r_f)T}$, where $S = 1.0404$ is the spot rate (USD per CHF), $r_d = 0.0025$, $r_f = 0$, and $T = 0.25$.

To compute this:
$e^{(0.0025 \times 0.25)} = e^{0.000625} \approx 1.000625195$.
$F = 1.0404 \times 1.000625195 \approx 1.04105$.

The quoted forward rate of 1.0300 is below the theoretical rate of approximately 1.04105, creating an arbitrage opportunity:

- Borrow 1 CHF at the Swiss risk-free rate of 0% (so no interest cost).
- Convert it to USD at the spot rate, receiving 1.0404 USD.
- Invest the 1.0404 USD at the U.S. risk-free rate of 0.25% for 3 months, growing to $1.0404 \times e^{0.0025 \times 0.25} \approx 1.04105$ USD.
- Enter into a long forward contract to buy 1 CHF in 3 months at 1.0300 USD per CHF.
- At maturity, use 1.0300 USD from the investment to purchase 1 CHF via the forward and repay the CHF loan.
- The risk-free profit is approximately $1.04105 - 1.0300 = 0.01105$ USD per CHF.

If the forward exchange rate is 1.0500 (above the theoretical rate of approximately 1.04105), the arbitrage opportunity reverses:

- Borrow 1.0404 USD at the U.S. risk-free rate of 0.25%.
- Use it to buy 1 CHF at the spot rate.
- Invest the 1 CHF at the Swiss risk-free rate of 0% for 3 months (remains 1 CHF).
- Enter into a short forward contract to sell 1 CHF in 3 months at 1.0500 USD per CHF.
- At maturity, deliver the 1 CHF via the forward, receiving 1.0500 USD, and use this to repay the USD loan (which has grown to approximately 1.04105 USD).
- The risk-free profit is approximately $1.0500 - 1.04105 = 0.00895$ USD per CHF.